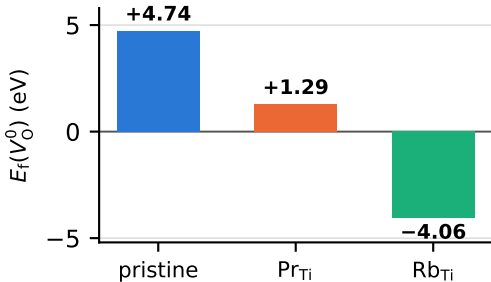


Anatase TiO_2 , PBE+U, $\mu_{\text{O}} = \frac{1}{2}E(\text{O}_2)$



Oxygen vacancies in Pr- and Rb-substituted anatase TiO_2

Both dopants lower the vacancy cost; Pr stays uphill, Rb goes downhill: a 5.36 eV contrast.

The order follows the formal hole count (0, 1, 3 per cell) and the cation size; the spin state of the Pr hole depends on $U(\text{O-2p})$.

Common supercell, U and O_2 reference